

UNIMODAL UNICYCLIC GRAPHS WITH SEVERAL BREAKS OF LOG-CONCAVITY

ABSTRACT. For a graph G write $I(G; x) = \sum_{k \geq 0} i_k x^k$, where i_k is the number of independent k -sets of G , and call an index $1 \leq k \leq \alpha(G) - 1$ a *break* of G if $i_k^2 < i_{k-1}i_{k+1}$. Levit and Kadrawi [4, Problem 6.6] ask whether one-edge closure of a multi-break or pattern-graph tree can produce a *unicyclic* graph whose independence polynomial is unimodal and has more than one break, or even several *consecutive* breaks. We exhibit unimodal unicyclic graphs of both kinds. The tree $T_{G,2,6}$ closed at one nonedge gives a unimodal unicyclic graph on 82 vertices with breaks exactly $\{40, 42\}$, and a pattern graph on 58 vertices closed at one nonedge gives breaks exactly $\{28, 29\}$, two *consecutive* breaks. One closure of $T_{G,7,6}$, on 282 vertices, is unimodal with *seven* breaks while its tree has six, so one-edge closure can also increase the number of breaks; in the five other witnesses below, including the two just named, the closure preserves the break set of the tree it closes. All arithmetic is exact integer arithmetic.

1. THE PROBLEM

Let G be a finite simple graph, $i_k = i_k(G)$ the number of independent sets of G of size k , $\alpha(G) = \max\{k : i_k > 0\}$ the independence number, and

$$I(G; x) = \sum_{k=0}^{\alpha(G)} i_k x^k$$

the independence polynomial. The sequence $(i_k)_{k=0}^{\alpha}$ is *unimodal* if it is non-decreasing up to some index and nonincreasing thereafter, and *log-concave* if $i_k^2 \geq i_{k-1}i_{k+1}$ for $1 \leq k \leq \alpha - 1$. We call an index k in that range a *break* of G when the inequality fails,

$$(1) \quad i_k^2 < i_{k-1}i_{k+1},$$

and write $B(G) \subseteq \{1, \dots, \alpha(G) - 1\}$ for the set of breaks. A graph is *unicyclic* if it is connected and $|E| = |V|$; a *one-edge closure* of a tree T is $T + e$ for a nonedge e of T , which is exactly a unicyclic graph.

The question addressed here is Problem 6.6 of Levit and Kadrawi [4].

Problem (multi-break unicyclic families). Starting from Bautista–Ramos’s multi-break families [1] and from the newer pattern-graph families with two and three consecutive breaks [2], decide whether one-edge closure

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can produce unicyclic graphs whose independence polynomials fail log-concavity at more than one index, or even at several consecutive indices, while remaining unimodal.

Nothing below is a tree-side result: the trees we close come from the families and the calculus of [1, 2], and the tree break sets we quote are recomputed only as baselines. The graph H_1 below is one of the closures of $T_{G,2,6}$ already computed in [4]; what we record for it, and for the other five witnesses, is the number and the position of the breaks. The theorem below is a statement about six explicit graphs and does not depend on how the problem above is read.

2. THE GRAPHS

We use the pattern-graph calculus of Bautista–Ramos, Guillén-Galván and Gómez-Salgado [2]. All graphs are rooted; the root of X is written $r(X)$.

- P_t is the path on t vertices, rooted at a leaf.
- $Z_k(H)$ is a new vertex v_0 together with k disjoint copies of H , with v_0 joined to the root of each copy; $r(Z_k(H)) = v_0$.
- $(G : H)_k$ is the disjoint union of G and $Z_k(H)$ with the edge $r(G) v_0$ added; the root stays $r(G)$.
- $(G : H)_k^{(m)}$ is the result of applying $(\cdot : H)_k$ to G m times in succession, the root staying $r(G)$ throughout. Hence

$$(2) \quad |(G : H)_k^{(m)}| = |G| + m(1 + k|H|).$$

- $S_{2,t} = (P_1 : P_1)_1^{(t)}$: a centre with t pendant paths of length two; $1 + 2t$ vertices.
- $T_{1,l} = (P_1 : P_2)_l^{(1)}$: a root v joined to a vertex v_0 that carries l pendant paths of length two; $2 + 2l$ vertices.
- $T_{G,m,t} = (P_2 : S_{2,t})_3^{(m)}$, the Bautista–Ramos multi-break tree (written $GT_{m,t}$ in [1, 2]): a root v_0 with one extra leaf and m pendant blocks, each block a hub joined to the centres of three copies of $S_{2,t}$. By (2), $|T_{G,m,t}| = m(4 + 6t) + 2$.

Labelling. Because our witnesses are named by the integer pair of their closure edge, we fix the labelling that the construction induces: vertices are numbered $0, 1, 2, \dots$ in the order in which the recipe above creates them. Concretely, in $(G : H)_k^{(m)}$ the vertices of G come first, in G 's own order (P_t being labelled $0, 1, \dots, t-1$ along the path from its root); then the m blocks in order, each block contributing its hub first and then its k copies of H in order, each copy in H 's own order. In $S_{2,t}$, for instance, this makes 0 the centre and the j -th pendant path the pair $(2j-1, 2j)$ for $1 \leq j \leq t$, with $2j-1$ the middle vertex and $2j$ the leaf. Every closure edge below is also described in words, so no reader has to trust the numbering.

Thus, explicitly:

- $H_1 = T_{G,2,6} + \{0, 3\}$, joining the root $v_0 = 0$ to the centre 3 of the first copy of $S_{2,6}$ in the first block, whose hub is 2; the unique cycle is the triangle 0 2 3.
- $H^* = (P_1 : S_{2,4})_2^{(3)} + \{0, 3\}$, joining the root $v = 0$ to the vertex 3, a middle vertex of a pendant path of the centre $c = 2$ of the first copy of $S_{2,4}$ in the first block, whose hub is $u = 1$; the unique cycle is the 4-cycle 0 1 2 3.
- $H^{**} = (P_3 : S_{2,4})_2^{(8)} + \{0, 5\}$, joining the root $v = 0$ of the base path 0 1 2 to the middle vertex 5 of a pendant path of the centre 4, whose hub is 3; the unique cycle is the 4-cycle 0 3 4 5.
- $H_A = (T_{1,3} : S_{2,4})_2^{(16)} + \{0, 10\}$, joining the root 0 of the base $T_{1,3}$ (labelled $0, \dots, 7$) to the middle vertex 10 of a pendant path of the centre 9, whose hub is 8; the unique cycle is the 4-cycle 0 8 9 10.
- $H_B = (T_{1,7} : S_{2,5})_2^{(13)} + \{1, 72\}$, joining the vertex 1 of the base $T_{1,7}$ (labelled $0, \dots, 15$; 1 is the neighbour of the root) to the middle vertex 72 of a pendant path of the centre 63 in the third block, whose hub is 62; the unique cycle is the 5-cycle 1 0 62 63 72.
- $H_7 = T_{G,7,6} + \{0, 3\}$, the same closure edge as H_1 one block-count higher; the unique cycle is the triangle 0 2 3.

3. THE WITNESSES

Theorem 1. *Each of the following six graphs is unicyclic, its independence polynomial is unimodal with a unique mode, and its set of breaks is exactly the set listed — not merely a subset of it.*

	H	$ V = E $	$\alpha(H)$	mode	$B(H)$
(a)	H_1	82	43	i_{25}	$\{40, 42\}$
(b)	H^*	58	31	i_{18}	$\{28, 29\}$
(c)	H^{**}	155	82	i_{47}	$\{78, 79, 80\}$
(d)	H_A	312	164	i_{95}	$\{162, 163\}$
(e)	H_B	315	164	i_{97}	$\{161, 162, 163\}$
(f)	H_7	282	148	i_{88}	$\{135, 137, 139, 141, 143, 145, 147\}$

Corollary 2. *One-edge closure produces unimodal unicyclic graphs whose independence polynomials fail log-concavity at more than one index — at two indices in (a), and at seven in (f) — and at several consecutive indices — two in (b) and three in (c) and (e). Here (d) and (e) are closures of $(T_{1,3} : S_{2,4})_2^{(16)}$ and $(T_{1,7} : S_{2,5})_2^{(13)}$, members of the two pattern-graph families of [2], and (a) and (f) closures of Bautista–Ramos’s $GT_{m,t}$ [1].*

The 58-vertex witness, in full. H^* has 58 vertices, so its whole independence polynomial fits on the page and Theorem 1(b) can be checked with a calculator. Its coefficients $i_0, \dots, i_{31}$ are

1, 58, 1595, 27709, 341818, 3190686, 23448526, 139359013, 682663978,
 2794877442, 9661712907, 28414927226, 71476627753, 154326951878,
 286560418561, 457792570649, 628535097664, 739676512678, 742815300664,
 632430324122, 452340586723, 268386668712, 129808551984, 49924871616,
 14717851602, 3137062048, 435476016, 31145353, 415590, 6304, 113, 1

The sequence increases strictly up to $i_{18} = 742815300664$ and decreases strictly afterwards, so it is unimodal with a unique mode. Four multiplications settle the break set at the top:

$$\begin{aligned} k = 27 : i_{27}^2 = 970033013494609 &\geq i_{26}i_{28} = 180979477489440 \quad (\text{holds}), \\ k = 28 : i_{28}^2 = 172715048100 &< i_{27}i_{29} = 196340305312 \quad (\text{break}), \\ k = 29 : i_{29}^2 = 39740416 &< i_{28}i_{30} = 46961670 \quad (\text{break}), \\ k = 30 : i_{30}^2 = 12769 &\geq i_{29}i_{31} = 6304 \quad (\text{holds}). \end{aligned}$$

The remaining 26 inequalities $1 \leq k \leq 26$ hold; they are large but immediate from the list. So $B(H^*) = \{28, 29\}$, two consecutive breaks.

In graph6 format H^* is the concatenation of the following five lines (no spaces, 277 characters, single-byte header since $58 \leq 62$):

```
y1C_H?QG??g??@??_?_?@?@???G?G???E?????G???_??@???C????_
??A????@???C?????G??@?????C????@?????_?????_????0?????@
?????_?????K?????????????C?????G?????G?????0?????@??
?????A????????_?????@????????@?????C?????????G????????@??
?????0?????????G????????A?????????C????????@?????????G
```

The decisive coefficients of the other five. In each case the full coefficient list is determined by the construction of §2; we print the coefficients that decide the break set, and the neighbouring index at which log-concavity holds, so that each entry of Theorem 1 can be confirmed without recomputing the whole polynomial.

H_1 , $\alpha = 43$: $i_{39} = 78555205917$, $i_{40} = 125984042$, $i_{41} = 622042$, $i_{42} = 485$, $i_{43} = 1$. Then

$$\begin{aligned} i_{40}^2 &= 15871978838657764 < i_{39}i_{41} = 48864637399022514, \\ i_{42}^2 &= 235225 < i_{41}i_{43} = 622042, \end{aligned}$$

while at the index between them log-concavity holds, $i_{41}^2 = 386936249764 \geq i_{40}i_{42} = 61102260370$: the two breaks of H_1 are *not* consecutive.

H^{**} , $\alpha = 82$: $i_{77} = 1135175576719$, $i_{78} = 3247342473$, $i_{79} = 9297742$, $i_{80} = 51245$, $i_{81} = 315$, $i_{82} = 1$. Then

$$\begin{aligned} i_{78}^2 &= 10545233136949755729, & i_{77}i_{79} &= 10554569637034468498, \\ i_{79}^2 &= 86448006298564, & i_{78}i_{80} &= 166410065028885, \\ i_{80}^2 &= 2626050025, & i_{79}i_{81} &= 2928788730, \end{aligned}$$

and in each of the three rows the left-hand value is the smaller, so 78, 79 and 80 are breaks, while $i_{81}^2 = 99225 \geq i_{80}i_{82} = 51245$ holds. The first of the three is decided by a margin of 0.088%, which is why the arithmetic here is exact integer arithmetic and never floating point.

H_A , $\alpha = 164$: $i_{160} = 14399685489601$, $i_{161} = 15059609304$, $i_{162} = 12154773$, $i_{163} = 9815$, $i_{164} = 9$. Then

$$\begin{aligned} i_{162}^2 &= 147738506681529 < i_{161}i_{163} = 147810065318760, \\ i_{163}^2 &= 96334225 < i_{162}i_{164} = 109392957, \end{aligned}$$

while at the index below, log-concavity holds:

$$i_{161}^2 = 226791832389123364416, \quad i_{160}i_{162} = 175024908397494015573.$$

H_B , $\alpha = 164$: $i_{159} = 1836675801123505133$, $i_{160} = 946417052360718$, $i_{161} = 367642616716$, $i_{162} = 146769284$, $i_{163} = 137465$, $i_{164} = 129$. Then

$$\begin{aligned} i_{161}^2 &= 135161093625787682624656, & i_{160}i_{162} &= 138904953140373090585912, \\ i_{162}^2 &= 21541222725872656, & i_{161}i_{163} &= 50537992306864940, \\ i_{163}^2 &= 18896626225, & i_{162}i_{164} &= 18933237636, \end{aligned}$$

and again the left-hand value is smaller in each of the three rows, so 161, 162 and 163 are breaks, while $i_{160}^2 \geq i_{159}i_{161}$ holds. Three consecutive breaks, inside one of the two pattern-graph families of [2].

H_7 , $\alpha = 148$. Its top fifteen coefficients $i_{134}, \dots, i_{148}$ are

176034211309656718484117918890005878151
 966322839077538885713606551341161815
 5610322511779111055524711146550627
 19499704262596590695297758206411
 69973942869517833994033432381
 174438847810507838032800383
 444508352733368700464454
 822213872739649589233
 1559903682459718229
 2122169611386407
 3022643280657
 2853892767
 2959617
 1535
 1

and the seven inequalities (1) at $k = 135, 137, 139, 141, 143, 145, 147$ all fail while the six at $k = 136, 138, 140, 142, 144, 146$ hold, giving alternating breaks $B(H_7) = \{135, 137, \dots, 147\}$.

4. PROOF OF THEOREM 1

Everything reduces to computing one independence polynomial per graph, and for a unicyclic graph that is a two-line reduction to forests.

Lemma 3. (i) *If F is a forest with components $F_1, \dots, F_c$ then $I(F; x) = \prod_j I(F_j; x)$, and for a tree rooted at r the two polynomials A_v (independent sets of the subtree at v containing v) and B_v (those not containing v) satisfy*

$$A_v = x \prod_{w \text{ child of } v} B_w, \quad B_v = \prod_{w \text{ child of } v} (A_w + B_w), \quad I = A_r + B_r.$$

(ii) If H is unicyclic and uv is an edge of its unique cycle, then $H - uv$ is a tree and

$$(3) \quad I(H; x) = I(H - uv; x) - x^2 I((H - uv) - N[u] - N[v]; x),$$

both graphs on the right being forests. (iii) If w lies on the unique cycle of H , then

$$(4) \quad I(H; x) = I(H - w; x) + x I(H - N[w]; x),$$

and again both graphs on the right are forests.

Proof. (i) is the standard rooted recursion: an independent set of the subtree at v either contains v , and then contains no child of v , or does not. (ii): independent sets of H are exactly the independent sets of $H - uv$ that do not contain both u and v ; those containing both are the sets $\{u, v\} \cup S$ with S independent in $(H - uv) - N[u] - N[v]$, whence (3). Deleting a cycle edge of a unicyclic graph leaves a spanning tree, and deleting further vertices leaves a forest. (iii) is the same dichotomy applied to w ; deleting a vertex of the unique cycle destroys it, so $H - w$ is a forest, and $H - N[w]$ is an induced subgraph of it. $\square$

Proof of Theorem 1. Each H in the table is $T + e$ for the tree T and nonedge e named in §2. A tree on n vertices has $n - 1$ edges, so $|E(H)| = |V(H)|$, and adding an edge to a connected graph keeps it connected: H is unicyclic. The vertex counts follow from (2): $|T_{G,2,6}| = 2(4 + 36) + 2 = 82$ and $|T_{G,7,6}| = 7 \cdot 40 + 2 = 282$; $|(P_1 : S_{2,4})_2^{(3)}| = 1 + 3(1 + 2 \cdot 9) = 58$; $|(P_3 : S_{2,4})_2^{(8)}| = 3 + 8 \cdot 19 = 155$; $|(T_{1,3} : S_{2,4})_2^{(16)}| = 8 + 16 \cdot 19 = 312$; $|(T_{1,7} : S_{2,5})_2^{(13)}| = 16 + 13 \cdot 23 = 315$. In each case the two ends of e are joined in T by the path displayed in §2, so the unique cycle is as stated.

For the polynomial, apply Lemma 3: (3) with uv the closure edge itself expresses $I(H; x)$ through two forest polynomials, each computed by the rooted recursion (i) in exact integer arithmetic. Then $\alpha(H) = \deg I(H; x)$; unimodality and the mode are read off the coefficient list; and each break is the single integer comparison (1). For H^* the resulting coefficient list and the four decisive comparisons are printed in §3 in full, so Theorem 1(b) is verified on the page. For the remaining five graphs the decisive comparisons are printed in §3, and *every* index $1 \leq k \leq \alpha - 1$ was checked; the accompanying program recomputes all six polynomials from the definitions of §2 by the two independent routes (3) and (4) and confirms that the break sets are exactly those listed. $\square$

5. REMARKS

Remark. In (a)–(e) of Theorem 1 the closure preserves the break set of the underlying tree exactly, with α unchanged; in (f) it does not, since $B(T_{G,7,6}) = \{135, 137, 139, 141, 143, 145\}$, so there the closure adds the break at the index $\alpha(H_7) - 1 = 147$. We do not claim that the number of breaks of a unimodal unicyclic graph can be made arbitrarily large: seven, on 282

vertices, is the largest number exhibited here. No minimality is claimed for any of the six witnesses.

Remark. The trees underlying H^* and H^{**} are pattern graphs in the sense of [2] but are not rows of that paper’s table of named families. Read strictly, so that “the newer pattern-graph families” means those named families, the consecutive-break clause is carried by H_A and H_B , which do lie in them.

6. REPRODUCTION

§2 determines all six graphs from their parameters alone, and Lemma 3 determines the polynomials from the graphs. A program `verify.py` accompanies this note, with a record of a run in `verify.output.txt`; it needs only Python 3 and its standard library,

```
python3 verify.py
```

and exits 0 only if every check passes. It rebuilds each graph from the definitions of §2, computes each independence polynomial by both (3) and (4), re-derives every integer printed above — the orders, the independence numbers, the modes, the coefficient lists, each printed product and the break sets in full — decodes the `graph6` string of §3 and matches it against the constructed H^* , and reproduces as controls published integers of [1, 2, 3]. It does not read the source e-print, so the statement of the problem in §1 lies outside its scope.

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